AREA FORMULA

Basic Shapes:

- **Rectangle:** Area = length × width ($A = l \times w$)
- **Square:** Area = side × side ($A = s^2$)
- **Triangle:** Area = (1/2) × base × height ($A = 0.5 \times b \times h$)
- **Circle:** Area = π × radius² ($A = \pi r^2$)

Quadrilaterals:

- **Parallelogram:** Area = base × height ($A = b \times h$)
- **Rhombus:** Area = (1/2) × diagonal 1 × diagonal 2 ($A = 0.5 \times d1 \times d2$)

Trapezoid (or trapezium): Area = (1/2) × (base1 + base2) × height ($A = 0.5 \times (b1 + b2) \times h$)

Other Shapes:

- **Sector of a Circle:** Area = $(\theta/360) \times \pi r^2$ (where θ is the angle of the sector in degrees)
- **Ring (area between two concentric circles):** Area = $\pi(R^2 - r^2)$ (where R is the radius of the outer circle and r is the radius of the inner circle)
- **Equilateral Triangle:** Area = $(\sqrt{3} / 4) * \text{side}^2$

I. 1. Area of a rectangle = (Length x Breadth).

$\therefore$ Length = $\left(\frac{\text{Area}}{\text{Breadth}}\right)$ and Breadth = $\left(\frac{\text{Area}}{\text{Length}}\right)$.

II. 2. Perimeter of a rectangle = 2(Length + Breadth).

III. Area of a square = $(\text{side})^2 = \frac{1}{2}(\text{diagonal})^2$.

IV. Area of 4 walls of a room = 2 (Length + Breadth) x Height.

V. 1. Area of a triangle = $\frac{1}{2}$ x Base x Height.

2. Area of a triangle = $s(s-a)(s-b)(s-c)$

where a, b, c are the sides of the triangle and $s = \frac{1}{2}(a + b + c)$.

3. Area of an equilateral triangle = $\frac{3}{4}$ x $(\text{side})^2$.

4. Radius of incircle of an equilateral triangle of side a = $\frac{a}{23}$.

5. Radius of circumcircle of an equilateral triangle of side a = $\frac{a}{3}$.

6. Radius of incircle of a triangle of area Δ and semi-perimeter r = $\frac{\Delta}{s}$.

VI. 1. Area of parallelogram = (Base x Height).

2. Area of a rhombus = $\frac{1}{2}$ x (Product of diagonals).

3. Area of a trapezium = $\frac{1}{2}$ x (sum of parallel sides) x distance between them.

VII. 1. Area of a circle = πR^2, where R is the radius.

2. Circumference of a circle = $2\pi R$.

3. Length of an arc = $\frac{2\pi R\theta}{360}$, where θ is the central angle.

4. Area of a sector = $\frac{1}{2}$(arc x R) = $\frac{\pi R^2\theta}{360}$.

VIII. 1. Circumference of a semi-circle = πR.

2. Area of semi-circle = $\frac{\pi R^2}{2}$.

1. Rectangle:

- **Area:** Length * Breadth (A = l * b)
- **Perimeter:** 2 * (Length + Breadth) (P = 2 * (l + b))

2. Square:

- **Area:** Side * Side ($A = s^2$)
- **Perimeter:** 4 * Side (P = 4s)

3. Triangle:

- **Area:** (1/2) * Base * Height (A = 1/2 * b * h)
- **Perimeter:** Sum of all three sides (P = a + b + c)
- **Equilateral Triangle Area:** ($\sqrt{3}$ / 4) * side² (A = ($\sqrt{3}$ / 4) * s^2)
- **Equilateral Triangle Perimeter:** 3 * side (P = 3s)

4. Circle:

- **Area:** π * Radius² ($A = \pi r^2$)
- **Circumference:** 2 * π * Radius ($C = 2\pi r$)

5. Parallelogram:

- **Area:** Base * Height (A = b * h)
- **Perimeter:** 2 * (Side1 + Side2) (P = 2 * (a + b))

6. Trapezoid:

- **Area:** (1/2) * (Sum of parallel sides) * Height (A = 1/2 * (a + b) * h)
- **Perimeter:** Sum of all four sides (P = a + b + c + d)

7. Rhombus:

- **Area:** (1/2) * (Product of diagonals) (A = 1/2 * d1 * d2)
- **Perimeter:** 4 * side (P = 4s)

AVERAGE FORMULA

1. Basic Average Formula:

- **Average = (Sum of all values) / (Number of values)**
- If you have 'n' numbers (x_1, x_2, x_3, ... x_n), the average is: ($x_1 + x_2 + x_3 + ... + x_n$) / n.

2. Average Speed:

- When dealing with average speed, the formula changes slightly. If you travel a distance 'd' at speed 'v_1' and another distance 'd' at speed 'v_2', the average speed is: **Average Speed = (2 * v_1 * v_2) / (v_1 + v_2) *
- In general, for varying distances and speeds, the average speed is calculated as: Average Speed = Total Distance / Total Time.

3. Average of First 'n' Natural Numbers:

- The average of the first 'n' natural numbers (1, 2, 3, ... n) is given by: Average = (n + 1) / 2.

4. Average of First 'n' Even Numbers:

- The average of the first 'n' even numbers (2, 4, 6, ... 2n) is: Average = n + 1.

5. Average of First 'n' Odd Numbers:

- The average of the first 'n' odd numbers (1, 3, 5, ... (2n-1)) is: Average = n.

TRIGONOMETRY

- Sinθ= 1/ Cosecθ or Sin θ.Cosecθ= 1
- Cosθ= 1/Secθ or Cos θ.Secθ= 1
- Tanθ= 1/Cotθ or Tanθ.Cotθ= 1
- Sin (A+B)= SinA.CosB + CosA.SinB
- Sin (A-B)= SinA.CosB – CosA.SinB
- Cos (A+B)= CosA.CosB- SinA.SinB
- Cos (A-B)= CosA.CosB+SinA.SinB
- Tan (A+B)= (TanA + TanB)/ (1-TanA TanB)
- Tan (A-B)= (TanA- TanB)/ (1+TanA TanB)

Basic Trigonometry Identities

Tangent & Cotangent
Tanθ = Sinθ/cosθ
Cotθ = Cosθ/Sinθ

Reciprocal
Sinθ = 1/Cosecθ
Cosecθ= 1/Sinθ
Cosθ= 1/Secθ
Secθ = 1/Cosθ
Tanθ = 1/Cotθ
Cotθ = 1/Tanθ

Pythagorean

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \operatorname{cosec}^2$

Even Odd

- $\sin(-\theta) = -\sin\theta$
- $\cos(-\theta) = \cos\theta$
- $\sec(-\theta) = \sec\theta$
- $\operatorname{cosec}(-\theta) = -\operatorname{cosec}\theta$
- $\tan(-\theta) = \cot\theta$
- $\cot(-\theta) = \tan\theta$

Logarithm formula

- Definition: $a^x = b$ can be represented in logarithmic form as $\log_a b = x$
- $\log a = x$ means that $10^x = a$.
- $10^{\log a} = a$ (The basic logarithmic identity).
- $\log(ab) = \log a + \log b$, $a > 0$, $b > 0$
- $\log(a/b) = \log a - \log b$, $a > 0$, $b > 0$.
- $\log a^n = n(\log a)$ (Logarithm of a power).
- $\log_x y = \log_m y / \log_m x$ (Change of base rule).
- $\log_x y = 1 / \log_y x$.
- $\log_x 1 = 0$ ($x \neq 0, 1$).
- The natural numbers 1, 2, 3,.... are respectively the logarithms of 10, 100, 1000, ... to the base 10.
- The logarithm of "0" and negative numbers are not defined.
- $\log_b 1 = 0$ ($\because b^0 = 1$)
- $\log_b b = 1$ ($\because b^1 = b$)
- $y = \ln x \rightarrow e^y = x$
- $x = e^y = \rightarrow \ln x = y$
- $x = \ln e^x = e^{\ln x}$
- $e^{\log_b x} = x$
- $\log_b b^y = y$

Laws of Logarithm

- $\log_b MN = \log_b M + \log_b N$ (where b, M and N are positive real numbers and $b \neq 1$)
- $\log_b(M/N) = \log_b M - \log_b N$ (where b, M and N are positive real numbers and $b \neq 1$)
- $\log_b(M^c) = c \log_b M$ (where b, M and N are positive real numbers and $b \neq 1$ and c is any real number)
- $\log_b M = \log M/\log b = \ln M/\ln b = \log_k M/\log_k b$ (where b, M and k are positive real numbers and $b \neq 1$, $k \neq 1$)
- $\log_b a = 1/\log_a b$ (where b, and a are positive real numbers and $b \neq 1$, $a \neq 1$)
- If $\log_b M = \log_b N$, then $M = N$ (where b, M and N are positive real numbers and $b \neq 1$)

Euler's Number

A mathematical consonant e is the base of the natural logarithm, known as Euler's number. It is also known as Napier's consonant.

$$e = 1 + \frac{1}{1} + \frac{1}{1.2} + \frac{1}{1.2.3} + \frac{1}{1.2.3.4} + \frac{1}{1.2.3.4.5} + \ldots\ldots\ldots\ldots \approx 2.71828$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots\ldots$$

1) $\log_x x = 1$

2) $\log_a 1 = 0$

3) $a^{\log_a x} = x$

4) $\log_a x \dfrac{1}{\log_x a}$

5) $\log_a(x^P) = p\,(\log_a x)$

6) $\log_a x = \dfrac{\log_b x}{\log_b a} = \dfrac{\log x}{\log a}$ - - - (Change of base formula)

7) $\log_a \left[\dfrac{x}{y}\right] = \log_a x - \log_b y$ - - - - (Division Rule)

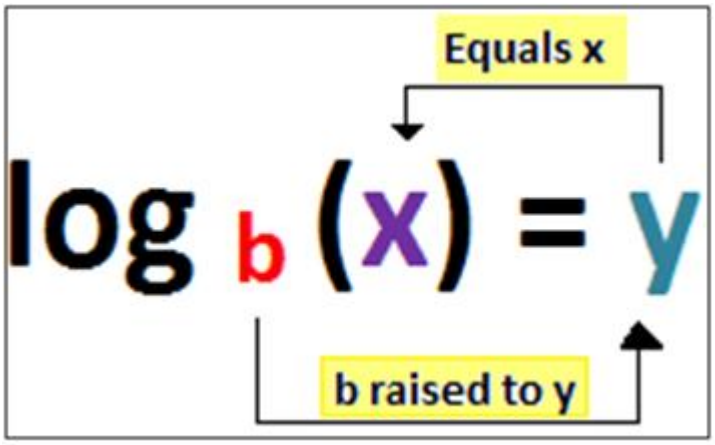
8) $\log_a (xy) = \log_a x + \log_b y$

Important points to remember

1) Logarithms are opposite to exponentials which means **logs are inverses of exponentials.**

2) Given logarithmic form can be converted into exponential form as shown below:

b = base y = exponent x = answer



Exponential form: $b^y = x$

Logarithm form: $\log_b (x) = y$

Log of numbers (2 – 10) (Must Remember)

Log 2 = 0.301
Log 3 = 0.477= 0.48
Log 4 = 0.60
Log 5 = 0.698 = 0.7
Log 6 = 0.778 = 0.78
Log 7 = 0.845 = 0.85
Log 8 = 0.90
Log 9 = 0.954= 0.96
Log 10 = 1

Geometry formula

The list of all the important Geometry Formulas is as given below

Name of the Shape	Formulas
Triangle	Area = ½ x base x perpendicular
	Perimeter = Sum of all sides
Rectangle	Area = Length x Breadth
	Perimeter = 2 (Length + Breadth)
	Diagonal (d) = √ ((length)2 + (breadth)2)
Square	Area = side x side
	Perimeter = 4 X Side
	Diagonal (d) = side x √2
Circle	Area = π x r x r
	Circumference = 2 x π x r
	Diameter = 2r
	where,
	r is the radius of the circle
Trapezium	Perimeter = Sum of all sides
	Area = ½ x sum of parallel sides x distance between the parallel sides
Parallelogram	Perimeter = Sum of all sides
	Area = base x height

Cube	Surface Area = 6 x (side)2
	Volume = (side)3
Cuboid	Surface Area = 2(lb + bh + hl)
	Volume = l x b h
	where,
	l = length of cuboid
	b = breadth of cuboid
	h = height of cuboid
Sphere	Surface Area = 4 x π x r2
	Volume = 4/3 x π x r3
	Diameter = 2r
	where,
	r is the radius of the sphere
Cone	Base Area = πr2
	Total Surface Area = πr (r + l)
	Volume = 1/3πr2h
	Slant height (l) = √ (h2 + r2)
	where,
	r = radius of cone
	h = height of the cone
	l = slant height of the cone
Cylinder	Total Surface Area = 2πr (r + h)
	Curved Surface Area = 2πrh
	Volume = πr2h

Base Area = πr2

where,

r = radius of the cylinder

h = height of the cylinder

Equation

Some Basic Formulae:

i. $(a + b)(a - b) = (a^2 - b^2)$
ii. $(a + b)^2 = (a^2 + b^2 + 2ab)$
iii. $(a - b)^2 = (a^2 + b^2 - 2ab)$
iv. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
v. $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
vi. $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
vii. $(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$
viii. When $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$.

Number Systems

- $1 + 2 + 3 + 4 + 5 + \ldots + n = n(n + 1)/2$
- $(1^2 + 2^2 + 3^2 + \ldots.. + n^2) = n (n + 1) (2n + 1) / 6$
- $(1^3 + 2^3 + 3^3 + \ldots.. + n^3) = (n(n + 1)/ 2)^2$
- Sum of first n odd numbers = n^2
- Sum of first n even numbers = $n (n + 1)$

Mathematical Formulas:

- $(a – b)^2 = (a^2 + b^2 – 2ab)$
- $(a + b)^2 = (a^2 + b^2 + 2ab)$
- $(a + b)(a – b) = (a^2 – b^2)$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
- $(a^3 – b^3) = (a – b)(a^2 + ab + b^2)$
- $(a^3 + b^3) = (a + b)(a^2 – ab + b^2)$
- $(a^3 + b^3 + c^3 – 3abc) = (a + b + c)(a^2 + b^2 + c^2 – ab – bc – ac)$
- When $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$
- $(a + b)^n = a^n + ({}^nC_1)a^{n-1} b + ({}^nC_2)a^{n-2} b^2 + \ldots + (nC_{n-1})ab^{n-1} + b^n$

Profit, Loss and Discount

- Profit (or) Gain = SP – CP
- Profit % = (Profit/CP) × 100

- SP = (100 + gain %)/100 × CP
- CP = 100/(100 + gain %) × SP
- Loss = CP – SP
- Loss % = Loss/(CP) × 100
- SP = (100 – loss %)/100 × CP
- CP = 100/(100 – loss %) × SP

LCM and HCF

LCM × HCF = Product of the numbers

LCM of co-prime numbers = Product of the numbers

Speed, Time and Distance

Distance = Speed × Time

Time = Distance/Speed

Speed= Distance/Time

Percentages

To find what percentage of x is y: (y/x) × 100

Increase N by S % = N(1+ S/100)

Decrease N by S % = N (1 – S/100)

Time and Work

If A can do a piece of work in n days, then A's 1 day's work = 1/n

If A's 1 day's work =1/n, then A can finish the work in n days.

Averages

Average = (Sum of observations/Number of observations)

Simple and Compound Interest

When interest is compounded Annually:

Amount=P(1+R100)n

When interest is compounded, Half-yearly:

Amount=P(1+R/2100)2n

When interest is compounded Quarterly:

Amount=P(1+R/4100)4n

When interest is compounded annually but time is in fraction, say 3 2/5 years:

Amount=P(1+R100)3×(1+25R100)

When rates are different for different years, say R_1%, R_2%, R_3% for 1st, 2nd and 3rd year, respectively.

Amount=P(1+R1100)+P(1+R2100)+P(1+R3100)

Present worth of Rs. x due n years hence is given by:

Present worth=x(1+R100)

Logarithm

- $\log_a (xy) = \log_a x + \log_a y$
- $\log_a (x/y) = \log_a x – \log_a y$
- $\log_x x = 1$
- $\log_a 1 = 0$
- $\log_a (xn) = n(\log_a x)$
- $\log_a x = 1/(\log_x a)$
- $\log_a x = (\log_b x)/(\log_b a) = (\log x)/(\log a)$

Probability

Sample Space: When we perform an experiment, then the set S of all possible outcomes is called the sample space.

Event: Any subset of a sample space is called an event.

The probability of Occurrence of an Event:

Let S be the sample and E be an event.

Therefore, **P(E) =n(E) / n(S)**

Surds and Indices

Laws of indices:

- $a^m \times a^n = a^{m+n}$
- $a^m/a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $(ab)^n = a^n b^n$
- $a^n/b^n = (a/b)^n$
- $a^0 = 1$
- $a^{-m} = 1/a^m$

Surds:

Let a be the rational number and n be a positive integer such that;

a1n=an
Here, an is called a surd of order n.

- an=a1n
- abn=an×bn
- abn=anbn
- (an)n=a
- anm=amn
- (an)m=amn

Set Theory & Function

De morgan's Law is the basic and most important formula for sets, which is defined as

(A ∩ B) ' = A' U B' and (A U B)' = A' ∩ B'

The relation R ⊂ A × A is said to be called as:

- Reflexive Relation: If aRa ∀ a ∈ A.
- Symmetric Relation: If aRb, then bRa ∀ a, b ∈ A.
- Transitive Relation: If aRb, bRc, then aRc ∀ a, b, c ∈ A.

If any relation R is reflexive, symmetric and transitive in a given set A, then that relation is known as an equivalence relation.

Permutation and Combination

Permutation Formula: A permutation is the choice of *r* things from a set of *n* things without replacement. Order matters in permutation.

$$nP_r = \frac{n!}{(n-r)!}$$

Combination Formula: A combination is the choice of *r* things from a set of *n* things without replacement. The order does not matter in combination.

$$nC_r = \frac{n!}{(n-r)!r!} = \frac{nPr}{r!}$$

Mixtures and Alligations

Alligation: It is the rule that enables us to find the ratio in which two or more ingredients at the given price must be mixed to produce a mixture of the desired price.

Mean Price: The cost of a unit quantity of the mixture is called the mean price.

Rule of Alligation:

If two ingredients are mixed, then

(Quantity of cheaper / Quantity of dearer) = (C.P. of dearer – Mean Price / Mean price – C.P. of cheaper)

Trigonometry

Trigonometric Identities:

- Sine = Opposite/Hypotenuse
- Secant = Hypotenuse/Adjacent
- Cosine = Adjacent/Hypotenuse
- Tangent = Opposite/Adjacent
- Co−Secant = Hypotenuse/Opposite
- Co−Tangent = Adjacent/Opposite

The reciprocal identities are given as:

- cosec θ = 1/sin θ
- sec θ = 1/ cos θ
- cot θ = 1/tan θ
- sin θ = 1/cosec θ
- cos θ = 1/sec θ
- tan θ = 1/cot θ

Coordinate Geometry

The Distance Between two Points A(x_1, y_1) and B(x_2, y_2):

$AB^2 = (x_2 – x_1)^2 + (y_2 – y_1)^2$

The Midpoint of a Line Joining Two Points

The midpoint of the line joining the points (x_1, y_1) and (x_2, y_2) is:

$[(x_1 + x_2)/1, (y_1 + y_2)/2]$

The Equation of a Line Using One Point and the Gradient

The equation of a line which has gradient m and which passes through the point (x_1, y_1) is:

$y - y_1 = m(x - x_1)$

Mensuration

Rectangle

Area = lb

Perimeter = 2(l+b)

Square

Area = $a \times a$

Perimeter = 4a

Triangle

Area = $(1/2) \times b \times h$

or

$\sqrt{[s(s - a)(s - b)(s - c)]}$, where $s = (a + b + c)/2$

Circle

Area = πr^2 or $\pi d^2/4$

Circumference = $2\pi r$ or πd

Area of sector of a circle = $(\theta \pi r^2)/360$

Cube

Volume: $V = a^3$

Lateral surface area = $4a^2$

Surface Area: $S = 6a^2$

Diagonal (d) = $\sqrt{3}$ a

Cuboid

Volume of cuboid: lbh

Total surface area = 2 (lb + bh + hl) or $6l^2$

Length of diagonal $=\sqrt{(l^2 + b^2 + h^2)}$

LOGARITHMS :

If am = x , then m = logax.
Properties :
1.log xx = 1
2.log x1 = 0
3.log a(xy) = log ax + log ay
4.log a(x/y) = log ax - log ay
5.log ax = 1/log xa
6.log a(xp) = p(log ax)
7.log ax = log bx/log ba
Note : Logarithms for base 1 does not exist.

AREA & PERIMETER :

Shape Area Perimeter
Circle ? (Radius)2 2?(Radius)
Square (side)2 4(side)
Rectangle length*breadth 2(length+breadth)
1.Area of a triangle = 1/2*Base*Height or
2.Area of a triangle = v (s(s-a(s-b)(s-c)) where a,b,c are the lengths of the sides and s = (a+b+c)/2
3.Area of a parallelogram = Base * Height
4.Area of a rhombus = 1/2(Product of diagonals)
5.Area of a trapezium = 1/2(Sum of parallel sides)(distance between the parallel sides)
6.Area of a quadrilateral = 1/2(diagonal)(Sum of sides)
7.Area of a regular hexagon = 6(v3/4)(side)2

VOLUME & SURFACE AREA :

Cube :
Let a be the length of each edge. Then,
1.Volume of the cube = a3 cubic units
2.Surface Area = 6a2 square units
3.Diagonal = v 3 a units
Cuboid :

Let l be the length, b be the breadth and h be the height of a cuboid. Then
1.Volume = lbh cu units
2.Surface Area = 2(lb+bh+lh) sq units
3.Diagonal = v (l2+b2+h2)
Cylinder :
Let radius of the base be r and height of the cylinder be h. Then,
2
1.Volume = PIr h cu units
2.Curved Surface Area = 2PIrh sq units
2
3.Total Surface Area = 2PIrh + 2PIr sq units
Cone :
Let r be the radius of base, h be the height, and l be the slant height of the cone. Then,
1.l2 = h2 + r2

2
2.Volume = 1/3(PIr h) cu units
3.Curved Surface Area = PIrl sq units
2
4.Total Surface Area = PIrl + PIr sq units
Sphere :
Let r be the radius of the sphere. Then,
3
1.Volume = (4/3)PIr cu units
2
2.Surface Area = 4PIr sq units
Prism :
Volume = (Area of base)(Height)
ALGEBRA :
1.Sum of first n natural numbers = n(n+1)/2
2.Sum of the squares of first n natural numbers = n(n+1)(2n+1)/6
3.Sum of the cubes of first n natural numbers = [n(n+1)/2]
4.Sum of first n natural odd numbers = n2
5.Average = (Sum of items)/Number of items

Arithmetic Progression (A.P.):

An A.P. is of the form a, a+d, a+2d, a+3d, ...
where a is called the 'first term' and d is called the 'common difference'
1.nth term of an A.P. tn = a + (n-1)d
2.Sum of the first n terms of an A.P. Sn = n/2[2a+(n-1)d] or Sn = n/2(first term + last term)

Geometrical Progression (G.P.):

A G.P. is of the form a, ar, ar2, ar3, ...
where a is called the 'first term' and r is called the 'common ratio'.
1.nth term of a G.P. tn = arn-1
2.Sum of the first n terms in a G.P. Sn = a|1-rn|/|1-r|

Permutations and Combinations :

1.nPr = n!/(n-r)!
2.nPn = n!
3.nP1 = n

1.nCr = n!/(r! (n-r)!)
2.nC1 = n
3.nC0 = 1 = nCn
4.nCr = nCn-r
5.nCr = nPr/r!

Tests of Divisibility :

1.A number is divisible by 2 if it is an even number.
2.A number is divisible by 3 if the sum of the digits is divisible by 3.
3.A number is divisible by 4 if the number formed by the last two digits is divisible by 4.
4.A number is divisible by 5 if the units digit is either 5 or 0.
5.A number is divisible by 6 if the number is divisible by both 2 and 3.
6.A number is divisible by 8 if the number formed by the last three digits is divisible by 8.
7.A number is divisible by 9 if the sum of the digits is divisible by 9.
8.A number is divisible by 10 if the units digit is 0.
9.A number is divisible by 11 if the difference of the sum of its digits at odd places and the sum of its digits at even places, is divisible by 11.

H.C.F and L.C.M :

H.C.F stands for Highest Common Factor. The other names for H.C.F are Greatest Common Divisor (G.C.D) and Greatest Common Measure (G.C.M).
The H.C.F. of two or more numbers is the greatest number that divides each one of them exactly.
The least number which is exactly divisible by each one of the given numbers is called their L.C.M.
Two numbers are said to be co-prime if their H.C.F. is 1.
H.C.F. of fractions = H.C.F. of numerators/L.C.M of denominators
L.C.M. of fractions = G.C.D. of numerators/H.C.F of denominators
Product of two numbers = Product of their H.C.F. and L.C.M.

PERCENTAGES :

1.If A is R% more than B, then B is less than A by R / (100+R) * 100
2.If A is R% less than B, then B is more than A by R / (100-R) * 100
3.If the price of a commodity increases by R%, then reduction in consumption, not to increase the expenditure is : R/(100+R)*100
4.If the price of a commodity decreases by R%, then the increase in consumption, not to decrease the expenditure is : R/(100-R)*100

PROFIT & LOSS :

1.Gain = Selling Price(S.P.) - Cost Price(C.P)

2.Loss = C.P. - S.P.
3.Gain % = Gain * 100 / C.P.
4.Loss % = Loss * 100 / C.P.
5.S.P. = (100+Gain%)/100*C.P.
6.S.P. = (100-Loss%)/100*C.P.

RATIO & PROPORTIONS:

1.The ratio a : b represents a fraction a/b. a is called antecedent and b is called consequent.
2.The equality of two different ratios is called proportion.
3.If a : b = c : d then a, b, c, d are in proportion. This is represented by a : b :: c : d.
4.In a : b = c : d, then we have a* d = b * c.
5.If a/b = c/d then (a + b) / (a – b) = (d + c) / (d – c).

TIME & WORK :

1.If A can do a piece of work in n days, then A's 1 day's work = 1/n
2.If A and B work together for n days, then (A+B)'s 1 days's work = 1/n
3.If A is twice as good workman as B, then ratio of work done by A and B = 2:1

PIPES & CISTERNS :

1.If a pipe can fill a tank in x hours, then part of tank filled in one hour = 1/x
2.If a pipe can empty a full tank in y hours, then part emptied in one hour = 1/y
3.If a pipe can fill a tank in x hours, and another pipe can empty the full tank in y hours, then on opening both the pipes,
the net part filled in 1 hour = (1/x-1/y) if y>x
the net part emptied in 1 hour = (1/y-1/x) if x>y

TIME & DISTANCE :

1.Distance = Speed * Time
2.1 km/hr = 5/18 m/sec
3.1 m/sec = 18/5 km/hr
4.Suppose a man covers a certain distance at x kmph and an equal distance at y kmph. Then, the average speed during the whole journey is 2xy/(x+y) kmph.

PROBLEMS ON TRAINS :

1.Time taken by a train x metres long in passing a signal post or a pole or a standing man is equal to the time taken by the train to cover x metres.
2.Time taken by a train x metres long in passing a stationary object of length y metres is equal to the time taken by the train to cover x+y metres.
3.Suppose two trains are moving in the same direction at u kmph and v kmph such that u>v, then their relative speed = u-v kmph.
4.If two trains of length x km and y km are moving in the same direction at u kmph and v kmph, where u>v, then time taken by the faster train to cross the slower train = (x+y)/(u-v) hours.
5.Suppose two trains are moving in opposite directions at u kmph and v kmph. Then, their relative speed = (u+v) kmph.
6.If two trains of length x km and y km are moving in the opposite directions at u kmph and v

kmph, then time taken by the trains to cross each other = (x+y)/(u+v)hours.
7.If two trains start at the same time from two points A and B towards each other and after crossing they take a and b hours in reaching B and A respectively, then A's speed : B's speed = (vb : v

SIMPLE & COMPOUND INTERESTS :

Let P be the principal, R be the interest rate percent per annum, and N be the time period.
1.Simple Interest = (P*N*R)/100
2.Compound Interest = P(1 + R/100)N – P
3.Amount = Principal + Interest